Appendix A: Main Experiment Details

A.1. Rebel on the Canal

A.1.1. Agents The simulation features two main types of agents: institutional actors (government and rebels) and residents.

Government and rebel organizations are structured hierarchically. Both government agents and rebel agents are categorized by rank into regular officials, secretary, and senior officials. Government agents belong to one of three policy factions—pro-canal, pro-maritime, or neutral—which influence their stance on infrastructure investment. All agents are assigned personality types with randomly selected words to modulate decision-making styles.

Each agent maintains both short-term and long-term memory. Short-term memory stores up to three recent dialogue entries. When this buffer fills, a large language model summarizes the conversation using the agent's identity and prior context, appending the result to a list. In addition to personal memory (e.g., past statements), agents have access to collective memory, which records annual policy decisions and their macro-level consequences, such as changes in employment rates, public satisfaction, and canal navigability.

Decision-making within each organization follows a four-step process. First, all members asynchronously submit initial opinions. Subsequent rounds involve iterative discussion informed by prior exchanges. An secretary then synthesizes these inputs into a shared information pool. Finally, designated leaders make binding decisions based on this consolidated view.

Government agents allocate annual resources across five dimensions: total public budget (an integer), the proportion of goods transported via canal (a float between 0 and 1), maintenance investment for the canal (integer), military support (integer), and tax adjustment (a float in [–0.1, +0.1]). Canal maintenance improves navigability, creates jobs for "canal maintenance workers," and reduces available fiscal resources. Rebel agents, by contrast, decide on propaganda budget, the scale of rebellion to stage, and target towns—either a single town or a list of towns with specified resource allocations per location.

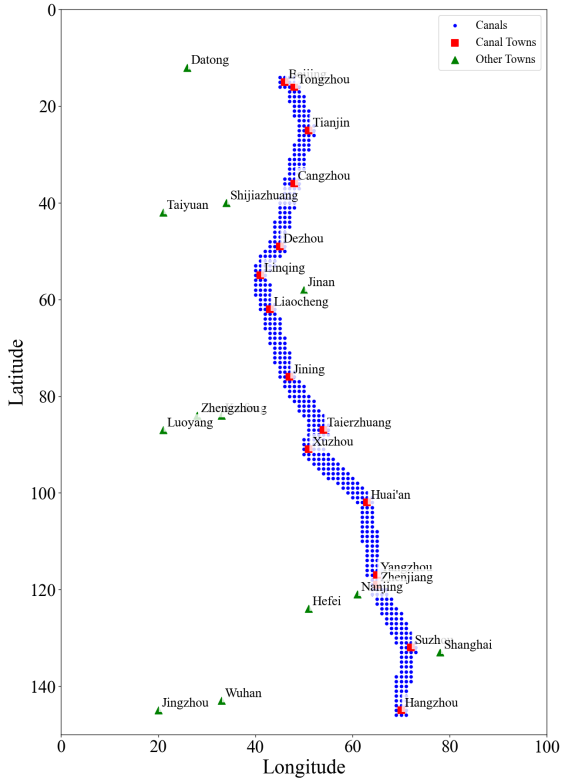
Resident agents represent the working-age population. Each has attributes including gender, remaining lifespan, employment status, job type, income, health index (1–5), geographic coordinates, assigned town, and a randomly generated two-word personality descriptor (e.g., "cautious and pragmatic"). Resident satisfaction is represented on a five-level qualitative scale: "hate the regime, vow to overthrow it"; "deeply resentful, denounce its incompetence"; "indifferent, disengaged from politics"; "tolerable, willing to comply"; and "loyal, ready to die for the state." This categorical measure directly influences the probability of joining a rebellion.

Residents possess only short-term memory (up to three interactions). Ordinary residents may choose to join a rebellion (more likely when dissatisfied), seek new employment if unemployed, migrate to another town upon job loss (guided by inter-town connectivity), update their satisfaction based on economic conditions and propaganda exposure, and post public remarks reflecting their mood. A subset of residents affiliated with rebels do not make active decisions but may post inflammatory statements with fixed probability.

A.1.2. Environment The environment consists of several interconnected subsystems.

The map system includes a two-dimensional river grid encoding the spatial layout of the Grand Canal, a scalar navigability index (ranging from 0 to 1) indicating current usability, a town connectivity graph in adjacency-list format for migration modeling, and a town attribute registry storing metadata such as name, coordinates, and town type (e.g.,

Figure 1 Map of the Grand Canal



canal-side or inland).The geographical distribution of the canal segments and the relative positions of the simulated towns (including canal-side and inland categories) are visualized in Figure 1.

Transport costs are dynamically computed. A global baseline transport cost scales linearly with total population. The canal transport cost is defined as the baseline cost multiplied by $(2 - \text{navigability index})$, ensuring it never falls below the baseline cost. Maritime transport is fixed at one-fifth of the baseline cost, creating an economic incentive to shift to sea routes as the canal degrades. Canal maintenance also directly affects the number of available infrastructure maintenance positions in the labor market.

Social relationships are modeled using two complementary structures. A heterogeneous graph captures identity-based ties such as friendships and professional connections, initialized with a power-law degree distribution $f(x) = cx^{-a}$. A hypergraph structure encodes community-based bonds: kinship clusters are formed via K-means clustering on resident coordinates (with minor stochastic noise), while hometown groups are defined by shared town membership. Agent i expresses opinions with a probability of d_i/d_{max}, where d_i and d_{max} denote the degree of agent i and the maximum degree of the network, respectively. Both types of graphs are updated during simulation, with edges stochastically rewired every three to five years.

Towns are represented as structured entities containing basic information (name, location, type), a registry of residents, a demographic management module, and a local employment market tracking labor dynamics. During initialization, the population is distributed proportionally across towns, with any remainder assigned to the primary town. Employment structures vary by town type; for instance, canal-side towns feature a higher density of transport-related jobs.

Each town's labor market includes six occupational categories: farmer, merchant, rebel, official/soldier, canal maintenance worker, and others. For each category, the system tracks total positions, employed residents (including wage levels), and base salary. Job availability responds dynamically to government investment and economic demand.

Finally, environmental and conflict dynamics are governed by simple mechanistic rules. Canal navigability decays annually according to the formula:

$$\phi_{t+1} = \max\big(0, \phi_t \cdot (1-\delta) - \gamma \cdot 0.6\big), \tag{1}$$

where ϕ represents the navigability index, δ is the natural decay rate, and γ is a climate impact factor. In battles between government and rebel forces, attrition rates depend on the force ratio $\alpha = G/R$. Letting $c = 0.1$ be the baseline attrition, $\alpha_{\max}$ be the threshold for extreme attrition, and $d \in [0.5, 1.0]$ a decay coefficient, government and rebel attrition probabilities are:

$$p_G = \begin{cases} \dfrac{c}{\alpha^d} + \epsilon & \text{if } \alpha \le \alpha_{\max} \\ 0.01 + \epsilon & \text{if } \alpha > \alpha_{\max} \end{cases} \tag{2}$$

$$p(R) = c \cdot \alpha^d + \epsilon, \tag{3}$$

with $\epsilon \sim \mathcal{U}(-0.03, 0.03)$. Actual losses are computed as these probabilities multiplied by respective force sizes. The actual attrition of government and rebel are $L_G = \lfloor p_G \cdot G \rfloor$ and $L_R = \lfloor p_R \cdot R \rfloor$, respectively.

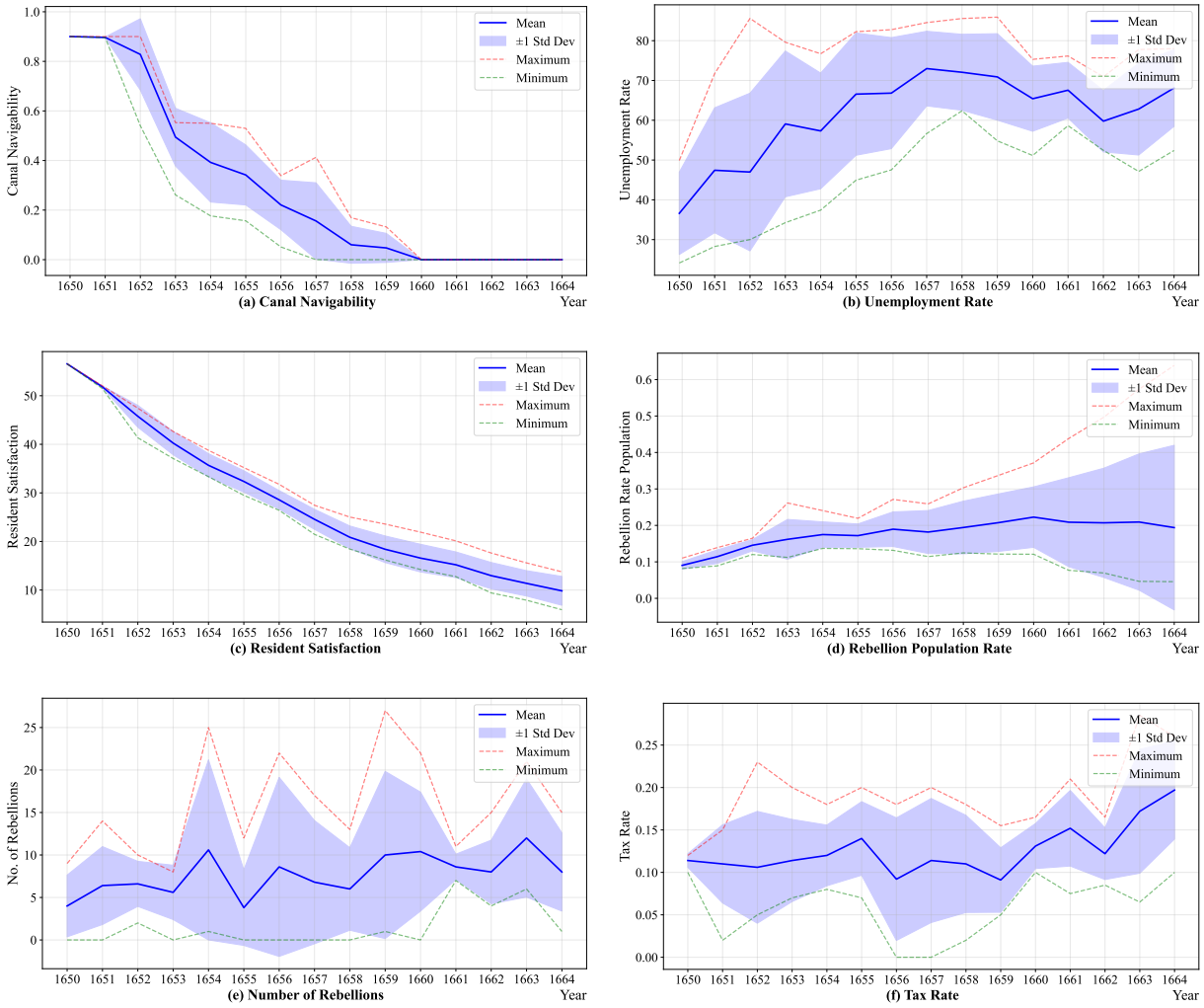
A.1.3. Complete Results of Canal Decay experiment Additional simulation results of the Canal Decay and Rebellion process are shown in Figure 2.

A.2. The Economic Origins of Government

A.2.1. Agents At the core of the simulation are two classes of agents: state officials and residents. State officials are characterized by two attributes: rank (ordinary official, secretary, or senior official) and personality type. The memory mechanism is same as that in Canal Decay experiment. Annually, officials collectively decide on two primary actions: (1) investment in canal maintenance, which directly improves navigability and thereby boosts urban incomes, and (2) adjustment of the tax rate, which influences both government revenue and citizen satisfaction.

Residents represent the labor-age population and are categorized as either rural farmers or urban citizens. Each resident possesses a profile that includes remaining lifespan, employment status (self-sufficient or integrated into a polity), occupation, income, satisfaction (a score from 0 to 100), health index (1–5), geographic location (x, y) and associated town, and a randomly assigned pair of personality descriptors. Like officials, residents maintain short-term memory of up to three recent interactions. Their annual decisions are also mediated by LLM-based reasoning. Rural farmers choose between joining a polity—thereby paying taxes but receiving stable income—or remaining self-sufficient, in which case their income is subject to climatic variability. Urban residents may elect to stay in their current city, migrate to another, or exit the polity entirely. All such choices update individual satisfaction and may trigger spatial relocation based on inter-town connectivity.

Figure 2 Complete Results of Canal Decay experiment



A.2.2. Environment The environment comprises a spatial map represented by several complementary data structures. A two-dimensional spatial layout matrix denotes the distribution of canals, while a scalar navigability index (ranging from 0 to 1) quantifies their transport efficiency. Inter-town migration is facilitated by a settlement connectivity graph, and comprehensive town metadata—including nomenclature, spatial coordinates, and settlement categories—is maintained in a centralized attribute registry.

Economically, income is dynamically calculated based on environmental and institutional parameters. Farmers' income is defined as a baseline wage modulated by an environmental impact coefficient, reflecting the susceptibility of agriculture to climatic variance. The income of urban residents is coupled with the state of canal infrastructure; residents in non-riparian settlements receive only 70% of the income relative to their riverside counterparts. Gross Domestic Product (GDP) is aggregated as the total income of all urban residents, serving as a proxy for economic productivity and state fiscal capacity.

A stochastic weather system introduces annual fluctuations via a climate impact factor (0–1), which concurrently diminishes agricultural yield and accelerates the natural degradation of canal infrastructure. The decay of the navigability index follows the same mechanism as defined in Canal Decay experiment. Strategic government investment in

maintenance can mitigate or reverse this deterioration, establishing a feedback loop between administrative action and economic stability.

Each simulation cycle follows a structured temporal sequence. Initially, the climate impact factor is sampled, and the navigability index is updated accordingly. Resident incomes are then recalculated relative to the revised environmental and infrastructural state. Macro-level indicators—including GDP, the government budget (derived from taxation and economic output), and the population birth rate (modeled as a function of aggregate satisfaction)—are updated next. The administrative phase follows, where officials engage in LLM-mediated deliberation to finalize fiscal and investment policies. During the agent behavior phase, individuals concurrently assess their survival status; those whose income falls below a subsistence threshold or who exceed their biological lifespan are removed from the system. Surviving agents make autonomous decisions regarding political affiliation and migration, potentially disseminating public statements through social networks to influence future collective behavior. Finally, each year concludes with a synchronization step: key performance metrics (population, GDP, satisfaction, etc.) are logged, and policy outcomes are archived into the governing agents' long-term memory to inform subsequent decision-making.

A.3. Information Delivery During India's Demonetization

The simulated population consists of autonomous agents representing residents, each endowed with basic demographic and psychological attributes including gender, remaining lifespan, a life satisfaction index (on a scale of 0–100), a health index (1–5), a geographical identifier representing a rural settlement, and a personality profile defined by latent behavioral traits. These attributes remain invariant throughout each experimental run but are re-initialized across strategy comparisons to ensure statistical independence.

A central feature of the model is the agents' dual memory system. Short-term memory records incoming information during each time step, including official messages (if any) and utterances received through social interactions. At the end of every time step, agents use a LLM to synthesize their short-term experiences into a coherent, stable representation of the policy event—this constitutes their long-term "knowledge memory." This distilled knowledge serves as the basis for subsequent decision-making and is directly queried during post-intervention assessments. Critically, all memory states and behavioral counters are fully reset before the initiation of each new dissemination strategy to maintain experimental purity.

Agents make decisions about whether to discuss the policy based on a context-aware planning process also mediated by an LLM. Their decision inputs include their personal state (e.g., personality, satisfaction), the content of any official message received, a "public notice" field that encodes meta-information about who else received the message (the key variable distinguishing common knowledge conditions), and their current knowledge memory. If an agent decides to speak, it generates natural-language commentary that is propagated through its social network connections.

The social network structure is fixed throughout each simulation run to isolate the effects of information strategy from network evolution. It combines heterogeneous pairwise ties (e.g., friendships, work relationships) with hyperedges representing group affiliations such as families or neighborhood clusters. Pairwise ties are generated via a power-law distribution to emulate real-world networks with hubs—individuals who have disproportionately many connections. Hyperedges are constructed using K-means clustering on agent attributes and spatial proximity, capturing dense local communities. When an agent speaks, its message is delivered to all immediate neighbors in this combined graph, triggering memory updates in recipients and enabling multi-step information diffusion.

Four distinct dissemination strategies are implemented, forming a 2×2 factorial design: two delivery modes (Broadcast vs. Seeding) crossed with two knowledge conditions (Common Knowledge vs. No Common Knowledge). In Broadcast (BC) strategies, every resident receives the full official message. In Seeding (S) strategies, only a small set of "seed" agents—selected based on a composite centrality measure combining degree, betweenness, and closeness—are given the message. The Common Knowledge (CK) manipulation concerns what agents are told about others' access to information. Under BC_CK, all agents receive the public notice: "You know that all villagers received the government message, and all villagers know that you received it." Under BC_NCK, no such notice is provided. In S_CK, seed agents are told: "Only some villagers received the message, and everyone knows you are one of them," while non-seeds hear: "Only some villagers received the message." In S_NCK, all agents—seeds and non-seeds alike—receive only the latter statement, obscuring who actually possesses the information.

Each strategy is executed over a fixed simulation period. Within each time step, the following sequence occurs: (i) the designated message and public notice are delivered according to the active strategy; (ii) every agent processes this input and decides whether to initiate a discussion; (iii) any generated speech propagates through the static social network, updating neighbors' short-term memories; and (iv) all agents summarize their accumulated inputs into updated long-term knowledge representations.

After the final time step of a given strategy, two standardized evaluations are conducted. First, a knowledge survey queries every agent on factual aspects of the demonetization policy; responses are parsed by an LLM to compute per-question and aggregate accuracy scores. Second, an incentivized choice task presents agents with a binary economic decision analogous to the field experiment: choosing between holding a now-invalid 500-rupee note (which can still be deposited in banks) or accepting a voucher redeemable in a few days for either 200 rupees or an equivalent value in lentils. The proportion selecting the rational option (the bankable note) serves as a behavioral proxy for policy comprehension and trust. Besides, the total volume of spontaneous policy-related discussions across the network during the intervention period is also recorded as a measure of social learning intensity.

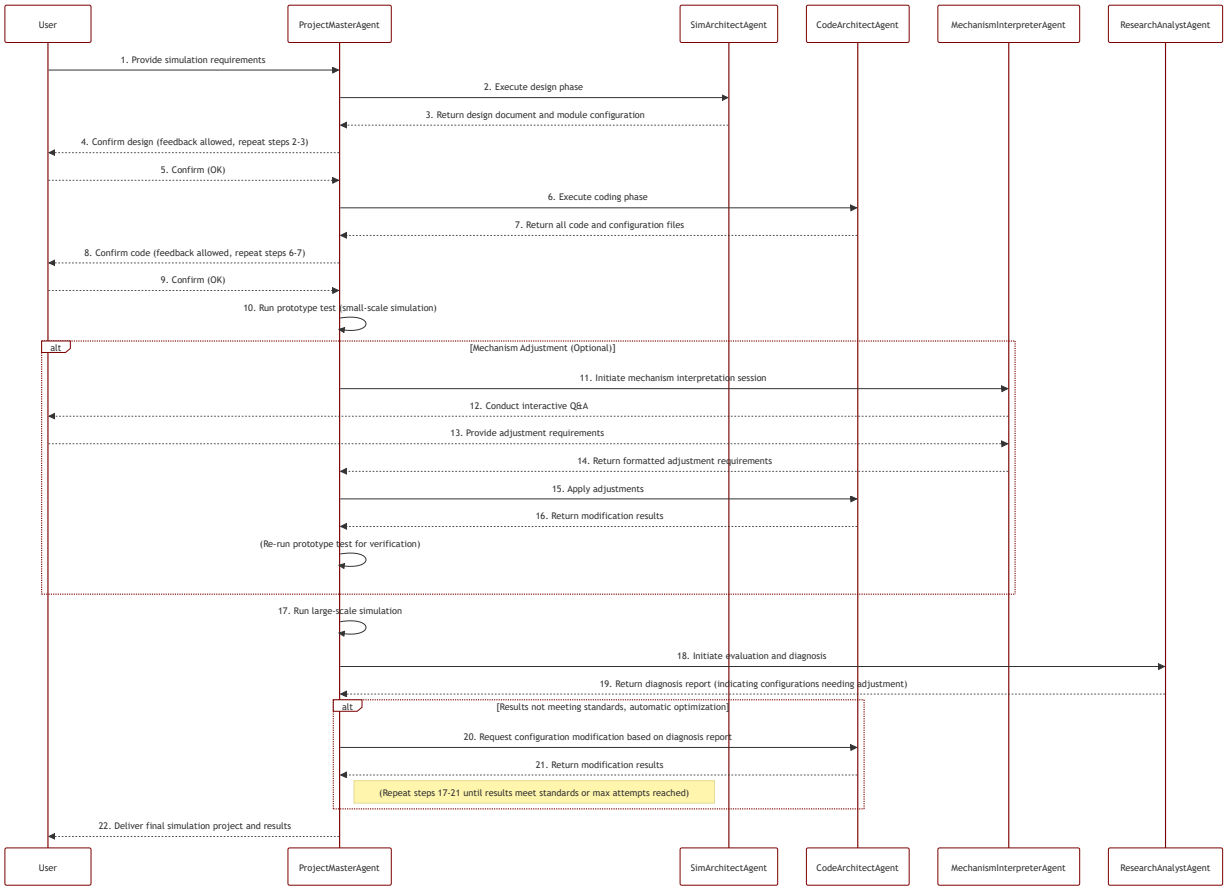
Appendix B: Auto-simulation Details

The auto-simulation pipeline in *Eco3S* relies on a set of specialized AI agents and structured data artifacts to enable end-to-end automation. The complete operational sequence—from requirement parsing and code synthesis to iterative optimization—is illustrated in the sequence diagram in Figure 3, which details the collaborative interactions between the four core AI agents.The following elaborates on the internal responsibilities of each agent and the semantic roles of the artifacts they produce or consume.

Agent Responsibilities

- **ProjectMasterAgent** serves as the central orchestrator. It parses user input into a structured requirement containing simulation name, description, and type (e.g., decision-based or survey-based). It then sequentially invokes other agents, manages execution of the generated simulation, captures runtime errors, and coordinates iterative refinement loops. In interactive mode, it pauses after design and coding phases to solicit human feedback.
- **SimArchitectAgent** acts as the system architect. Given the structured requirement, it generates a high-level design specification that defines agent types (e.g., residents, officials), environmental components (e.g., climate, geography, job market, heterogeneous information networks), interaction protocols (e.g., taxation, migration, rebellion triggers), and decision-making modes (individual vs. group consensus). It also produces a module activation plan indicating which functional subsystems should be included.

Figure 3 Workflow of the auto-simulation framework



- **CodeArchitectAgent** functions as a full-stack developer. It synthesizes executable Python code for the simulator core and program entrypoint based on the design specification and module plan. It also generates detailed runtime configuration files that govern simulation parameters, initial agent states, and LLM prompt templates for behavioral control. Crucially, it supports two forms of automated repair: (1) fixing runtime crashes by analyzing error traces and modifying code/configurations; (2) adjusting configurations in response to diagnostic feedback to improve result quality.
- **ResearchAnalystAgent** serves as the evaluation expert. After a successful simulation run, it compares the output data against the expected behaviors described in the original design. If outcomes deviate (e.g., rebellion rates are too low), it produces a diagnostic report that identifies the most likely faulty components (e.g., overly generous job market settings) and suggests concrete, actionable modifications to relevant configurations or agent prompts.

Artifact Types and Functions

The agents communicate through a set of well-defined artifact types:

- **Structured Requirement**: A machine-readable representation of the user's natural language input, enabling deterministic downstream processing.
- **System Design Specification**: A comprehensive document detailing the simulated entities, environment dynamics, interaction rules, and decision mechanisms. This serves as the single source of truth for implementation.

- **Module Activation Plan**: A declarative configuration that specifies which functional modules (e.g., job market, climate system) are enabled, acting as a high-level switchboard for code generation.
- **Executable Source Code**: The generated Python implementation, including the simulator logic and execution entrypoint. This code is fully functional without manual intervention.
- **Runtime Configuration Set**: A collection of structured files that define numerical parameters, initial states, and LLM prompts for agent behaviors. These decouple policy logic from program structure, enabling easy tuning and counterfactual experiments.
- **Simulation Output Data**: Structured records (e.g., time-series of agent actions, environmental states, macro indicators) produced during execution, used for evaluation.
- **Diagnostic Report**: A machine-readable analysis that links observed outcome deviations to specific configuration or behavioral flaws, providing precise guidance for iterative refinement.

Iterative Refinement and Error Handling The system supports two closed-loop mechanisms:

1. **Runtime Error Recovery**: When simulation execution fails, the ProjectMasterAgent captures the full error trace and instructs the CodeArchitectAgent to analyze the root cause. The latter identifies relevant code and configuration components, then applies targeted fixes—repeating until the simulation runs successfully.
2. **Result-Driven Optimization**: If the ResearchAnalystAgent deems results unsatisfactory, it generates a diagnostic report. The ProjectMasterAgent then initiates a refinement loop: the CodeArchitectAgent adjusts the runtime configuration (or prompts) based on the report, and the simulation is re-run. This cycle continues until results meet quality thresholds or a maximum iteration count is reached.

All artifacts for a given simulation are stored in an isolated directory, ensuring reproducibility and seamless integration with *Eco3S*'s counterfactual mechanism, which allows users to modify configurations and restart from any saved state.

Appendix C: More Experiments on Auto-simulation

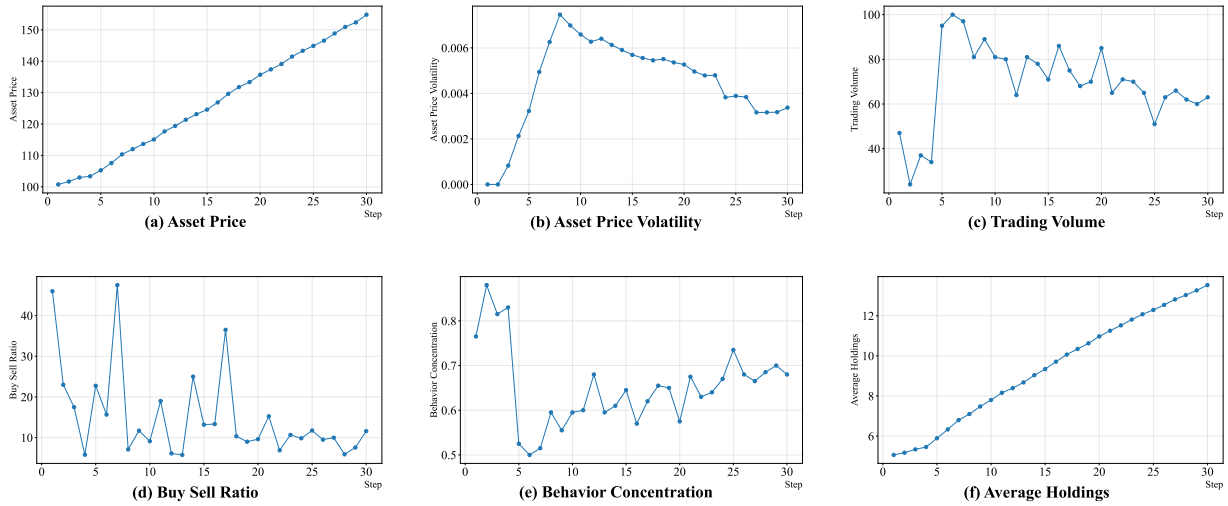
We demonstrate details regarding the auto-simulation experiments, including prompt, experiment results and auto-analysis results.

C.1. Herding Effect

The **herding effect simulation** models herding behavior in financial markets, where each agent represents an investor making buy, sell, or hold decisions based on market signals and social interactions. The core mechanisms include a dynamic social network for information diffusion, a feedback loop between trading activity and asset prices. The model runs over discrete time steps, allowing emergent market phenomena to arise from individual-level behaviors.

As shown in Figure 4, the results demonstrate clear manifestations of herding behavior. Asset prices rose significantly over the simulation period, accompanied by a steady increase in price volatility, from zero to around 0.0032, with peaks reaching 0.0074, indicating heightened market instability driven by synchronized trading. Trading volume remained consistently high, and the buy/sell ratio was heavily skewed toward buying (ranging from 5.75 to 47.50), reflecting strong "chasing-the-trend" dynamics. Behavioral concentration averaged 0.65, with multiple time points exceeding 0.83, confirming high decision homogeneity among agents. Additionally, average holdings increased by over 167%, further evidencing aggressive collective buying. Overall, the simulation successfully replicates key theoretical features of financial herding, including price amplification, elevated volatility and behavioral synchronization.

Figure 4 Results of Herding Effect Experiment



The model comprises four interconnected subsystems: the agent system, the social network, the market feedback mechanism, and the time–population dynamics. Each agent represents an autonomous investor endowed with initial cash, asset holdings, risk preference, and a satisfaction score that reflects their subjective evaluation of investment outcomes. At every discrete time step, agents decide whether to buy, sell, or hold the single tradable asset based on public market signals and opinions received through their social connections. When an agent executes a trade, it directly affects the market-clearing price. This new price then feeds back into agents' next-round evaluations, closing the loop between micro-behavior and macro-dynamics. Population size is not fixed; agents may exit the simulation if their satisfaction falls below a threshold.

The prompt is as following:

Prompt

Design an experiment to verify the herd effect in financial markets, one that can clearly observe herd phenomena.

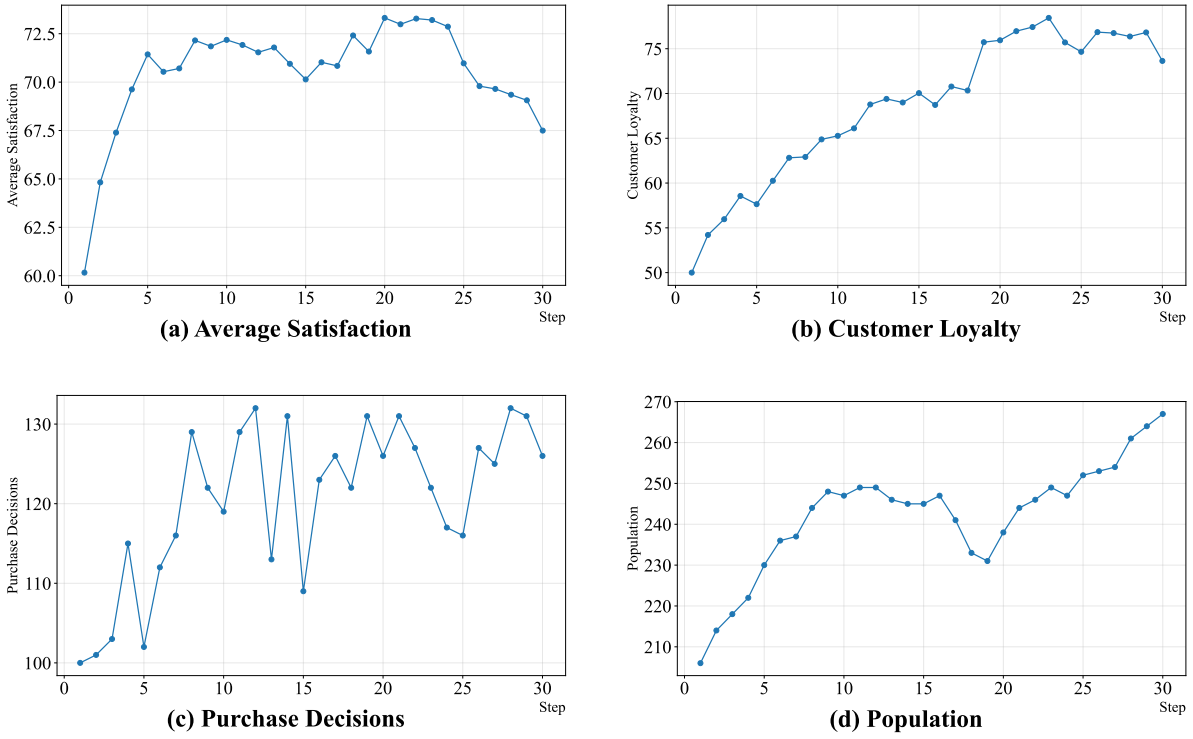
C.2. Custom Satisfaction and Loyalty

The **custom satisfaction and loyalty simulation** investigates the dynamic interplay among customer satisfaction, loyalty, and purchasing behavior within a service or product market. Using a multi-agent modeling approach, the system represents individual consumers as resident agents endowed with attributes such as income, employment status, and evolving satisfaction levels. The core mechanisms include word-of-mouth communication through a social network, where each agent's satisfaction influences peers' perceptions and decisions; a macroeconomic environment in which agents earn income through employment; and a demographic subsystem where population growth is modulated by average satisfaction, reflecting long-term social feedback. Agents make parallel decisions on consumption, work, and social interaction.

The experimental results demonstrate a strong positive correlation between satisfaction and loyalty, with average satisfaction rising from 60.16 to a peak of 73.32 between step 1 and step 20, accompanied by loyalty increasing from 50.00% to 75.94%. A slight decline in both metrics after step 24 coincides with population growth outpacing

service capacity, suggesting a dilution of per capita service quality. Purchase decisions track closely with satisfaction trends, validating the embedded behavioral rule that low-satisfaction agents are less likely to buy. Notably, loyalty exhibits hysteresis—declining more slowly than satisfaction—highlighting the "stickiness" of established customer relationships. The simulation successfully validates key theoretical propositions: (1) sustained high service quality drives a virtuous cycle of satisfaction, loyalty, and consumption; (2) a satisfaction threshold of 70 acts as a critical inflection point for loyalty formation; and (3) unmanaged scale expansion can undermine satisfaction, posing latent churn risks even when loyalty remains elevated. Overall, the model provides quantitative support for customer-centric strategies and underscores the necessity of aligning service capacity with market growth.

Figure 5 Results of Custom Satisfaction Experiment



The core of the model consists of resident agents, each representing an individual consumer endowed with attributes including income, employment status, and a time-varying satisfaction score derived from perceived service and product quality. Satisfaction directly influences both purchase probability and loyalty status, where loyalty is defined as a binary state triggered when an agent's satisfaction reaches or exceeds 70 on a 0–100 scale. Agents interact within a structured social network through which word-of-mouth communication propagates: positive or negative experiences are shared with connected peers, thereby adjusting their satisfaction levels in subsequent time steps. This mechanism captures the amplification effect of word-of-mouth on collective sentiment.

Each simulation year proceeds in four phases: (1) macro indicators (e.g., average satisfaction, loyalty ratio) are computed; (2) all agents make parallel decisions on employment, consumption, and social expression; (3) decisions are executed—consumption reduces disposable income, employment generates income, and expressed opinions propagate through the network; and (4) all metrics are logged for analysis.

Key behavioral rules include: (1) Agents with satisfaction ¡ 40 never purchase; those with satisfaction between 40 and 60 purchase with 30% probability; agents with satisfaction ≥ 60 purchase normally. (2) Loyalty is updated in each step based solely on current satisfaction. (3) Service quality is treated as an exogenous system-level parameter that uniformly elevates baseline satisfaction across all agents when improved, though in this baseline run, it remained constant.

The simulation was configured with an initial population of 206 agents, ran for 30 steps, and used a fixed birth rate modulated by average satisfaction. All data—including yearly values of population, unemployment rate, average satisfaction, customer loyalty, number of purchase decisions,

The prompt is as following:

Prompt

System Goal: Observe how dynamic changes in service quality and product experience impact customer satisfaction, loyalty, and purchasing decisions. Agent Types: Customers are the core simulation objects, each with attributes: perceived service quality (range [0,1]), product experience score (range [0,1]), satisfaction (calculated from service and product experience), loyalty (accumulated based on historical satisfaction), purchase probability (determined by loyalty and current satisfaction). Environment Description: Merchant system with baseline service quality and product quality as system-level variables influencing all customers' perceptions. The system operates on discrete time steps, each representing a standardized decision cycle over 10 periods.

C.3. Asset Bubble

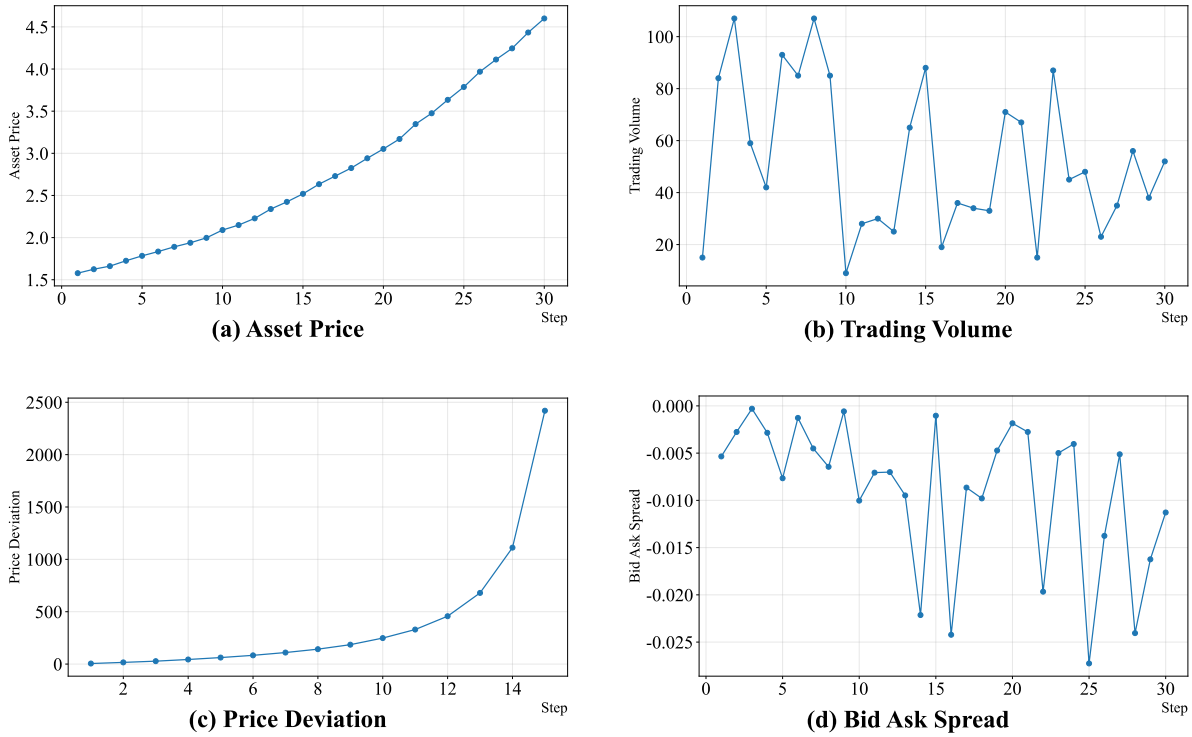
The model replicates the classic asset market design (**?**), where a synthetic asset pays a fixed dividend per period and expires worthless after a predetermined number of trading periods, thereby establishing a known fundamental value that declines linearly over time. The primary objective is to investigate how bubbles—defined as sustained deviations of market prices above fundamental value—emerge endogenously from the interactions of heterogeneous, boundedly rational agents.

The simulation features a population of trader agents endowed with diverse risk preferences (risk-averse, risk-neutral, risk-seeking) and trading strategies (momentum, value-based, or hybrid). Trading occurs via a continuous double auction mechanism: agents submit limit orders, which are matched by a central order book to determine transaction prices and volumes. Market outcomes—including price, volume, bid-ask spread, and price deviation from fundamental value—are recorded each period. The model also incorporates demographic dynamics, allowing new agents to enter the market based on satisfaction-driven birth rates, thereby introducing endogenous liquidity shocks.

Experimental results demonstrate a clear bubble formation process. Over 30 trading periods, the asset price rose from 1.58 to 4.60—a 191% increase—despite its fundamental value declining to zero after period 15. Price deviation from fundamentals peaked at over 2400% in period 15 and remained positive thereafter, indicating speculative trading detached entirely from fundamentals. Trading volume exhibited volatility with notable spikes, while the bid-ask spread turned negative and narrowed over time, reflecting persistent excess demand and strong buying pressure. The simulation successfully reproduces key behavioral phenomena—herding, momentum chasing, and speculative mania—and validates the hypothesis that heterogeneous, LLM-driven agents interacting in a double-auction market can generate robust bubble dynamics even in the presence of transparent fundamentals.

The experiment implements an agent-based computational model of a finite-horizon asset market. In this setting, a synthetic asset pays a fixed dividend per period for a predetermined number of trading periods (default: 30), after

Figure 6 Results of Asset Bubble Experiment



which it expires worthless. The fundamental value of the asset at any period t is thus precisely defined as the sum of all remaining dividends:

$$V_t = D \times (T - t), \tag{4}$$

where D is the per-period dividend amount and T is the total number of trading periods. This known and declining fundamental value serves as the benchmark against which price deviations—interpreted as bubbles—are measured.

The market consists of heterogeneous trader agents, each initialized with identical endowments of cash and asset units to ensure a level playing field. Agents differ along two key behavioral dimensions: (1) trading strategy, which includes momentum trading (chasing trends), value investing (anchoring to fundamental value), or a hybrid approach; and (2) risk preference, categorized as risk-averse, risk-neutral, or risk-seeking. At the beginning of each trading period, an agent receives a structured prompt containing contextual information—including current and historical prices, order book depth, remaining dividend periods, its own portfolio holdings, and observed price trends—and outputs a structured decision to either submit a buy order, a sell order, or hold.

Trading occurs via a continuous double auction mechanism mediated by a centralized order book. Buy orders are sorted in descending price order, and sell orders in ascending order. The matching algorithm iteratively pairs the highest bid with the lowest ask whenever the bid price is at least as high as the ask price. Transactions are executed at the midpoint of the matched prices, and partial fills are permitted. The market-clearing price for the period is computed as the volume-weighted average of all transaction prices; if no trades occur, the price is carried forward from the previous period.

At the start of each period, dividends are paid to all asset holders, increasing their cash balances and potentially influencing subsequent trading behavior. The simulation also incorporates demographic dynamics: new agents may

enter the market based on a satisfaction-driven birth rate, while existing agents may exit due to stochastic mortality. This endogenous population change introduces liquidity shocks and can amplify feedback loops between price movements and participant inflows.

The prompt is as following:

Prompt

I would like to build a double auction market simulation for finite-lived assets. Heterogeneous agents, driven by their risk preferences, trading strategies (fundamental analysis, technical analysis, herding behavior), and market beliefs, will make buy and sell decisions. The goal is to observe how, under the constraints of fixed dividend periods and maturity reset to zero, asset prices deviate from fundamental value to form boom and bust cycles. This simulation aims to verify the impact of order book matching mechanisms, speculative behavior, and herd effects on market irrational exuberance and price volatility.

C.4. Schelling Segregation Model

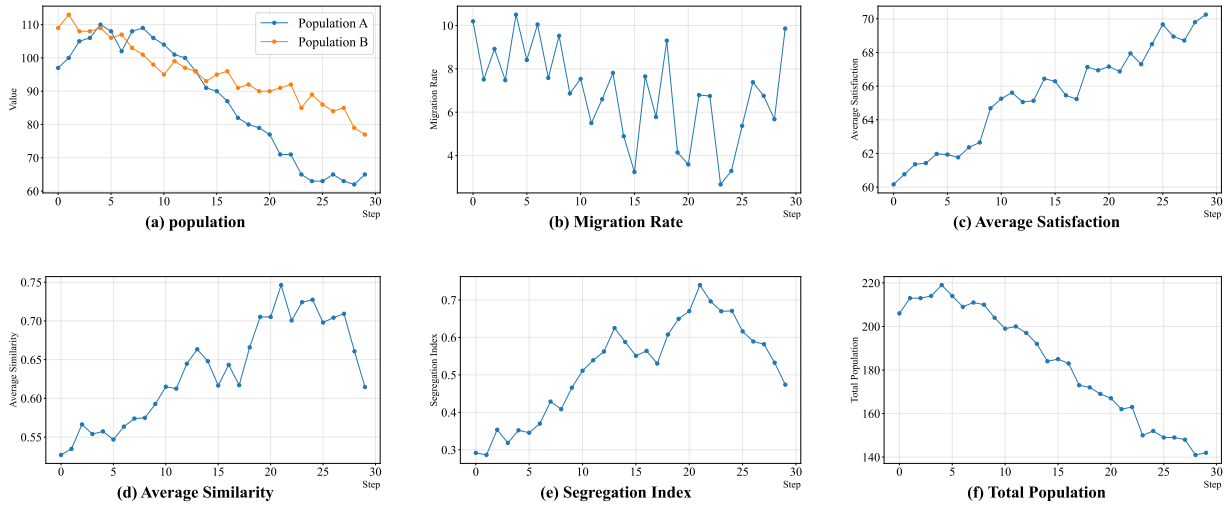
This study implements a computational reconstruction of Schelling's segregation model (**?**) to explore how macro-level spatial segregation emerges from micro-level individual preferences. The simulation features a grid-based virtual city populated by two types of autonomous agents (Type A and Type B), each with a fixed ethnic identity. Agents evaluate their satisfaction based on the proportion of similar neighbors within their Moore neighborhood (8 surrounding cells). If this similarity falls below a configurable threshold (e.g. 30%, 50%, or 80%), agents consider relocating. The model incorporates dynamic population processes—births influenced by average satisfaction, deaths tied to health and living costs—and updates social networks periodically. Over discrete annual time steps, the system tracks migration decisions (aided by LLM-based reasoning), population changes, and evolving spatial patterns, aiming to quantify how mild in-group preferences can still lead to pronounced segregation.

Results show that even with a moderate preference threshold of 50%, significant spatial segregation emerges: the segregation index rises from an initial 0.29 to 0.74, while average neighbor similarity increases from 0.53 to 0.75. Concurrently, average resident satisfaction improves from 60.16 to 66.87, illustrating the classic "Schelling paradox"—individuals become happier with their local environment precisely as society becomes more segregated. The population declines over the simulation due to economic attrition, but both groups shrink proportionally, preserving demographic balance. Migration rates fluctuate between 5.50% and 9.30%, reflecting ongoing spatial reorganization without convergence to a static equilibrium within the simulated timeframe. Overall, the experiment successfully validates Schelling's core insight: mild in-group preferences at the individual level can unintentionally generate pronounced patterns of macro-segregation through decentralized, self-organizing processes.

The environment is initialized with a population of 206 residents, equally distributed between two immutable types (A and B), randomly placed across a map of configurable dimensions (default: 30 × 30 cells) with approximately 10% of cells left vacant to enable mobility. Each resident agent evaluates its local neighborhood using the Moore definition (the eight adjacent cells) and computes a neighbor similarity ratio—the proportion of neighbors belonging to the same type. A fixed similarity threshold of 0.5 is applied as the baseline behavioral rule: if the observed similarity falls below this threshold, the agent expresses a desire to relocate.

Migration decisions are executed by moving the dissatisfied agent to a randomly selected vacant cell. The simulation progresses in discrete annual time steps over a total duration of 50 years. To introduce demographic realism, the model

Figure 7 Results of Schelling Model Experiment



incorporates dynamic population mechanisms: at each time step, new agents are probabilistically generated based on the current average satisfaction level of the population, while existing agents may be removed from the system if their economic resources fall below the basic living cost—a parameter set in the configuration file. This introduces a non-conservative population dynamic, leading to a net decline in total population over time.

In addition to spatial dynamics, the model maintains a social network layer represented as an undirected graph, where edges between agents are periodically rewired every 3–5 years to reflect evolving social ties. However, this network does not directly influence migration decisions in the current implementation and is retained for potential future extensions involving social influence or information diffusion.

At every time step, the simulation records seven key metrics: total population, migration rate (defined as the percentage of agents who moved in that year), average satisfaction (a linear function of neighbor similarity and affordability), average neighbor similarity across all agents, the index of dissimilarity as the primary measure of spatial segregation, and the respective counts of type-A and type-B residents. The index of dissimilarity is computed over the entire grid treated as a set of spatial units, following the standard formula:

$$D = \frac{1}{2} \sum_{i=1}^{n} \left| \frac{a_i}{A} - \frac{b_i}{B} \right|, \tag{5}$$

where a_i and b_i are the numbers of type-A and type-B residents in cell i, and A and B are the total populations of each type. All data are logged in CSV format with timestamps and process identifiers to support reproducibility and parallel experimentation.

The prompt is as following:

Prompt

Reproduce the Schelling Segregation Model. The Schelling Model is an economic term proposed by Nobel laureate Thomas Schelling in 1971. It uses a multi-agent model to reveal the emergent mechanism of urban ethnic spatial segregation. The model simulates urban population distribution based on an N×N grid, with two types of residents (e.g., red/blue) randomly distributed on a grid containing 10% empty cells. When the proportion of similar neighbors among the eight surrounding neighbors of a resident falls below a set threshold, they choose to move to any empty cell. The model includes two core mechanisms: individual behavioral rules based on thresholds and the emergence of macro patterns from micro-level interactions. Experiments show that when residents' clustering tendency is 50%, clear boundaries form; with a 30% clustering tendency, the average similarity still reaches 70%; while with an 80% clustering tendency, frequent moves lead to integration. This indicates a nonlinear relationship between individual preferences and group outcomes, known as the "emergence" phenomenon.

Appendix D: System Interface and Operational Workflow

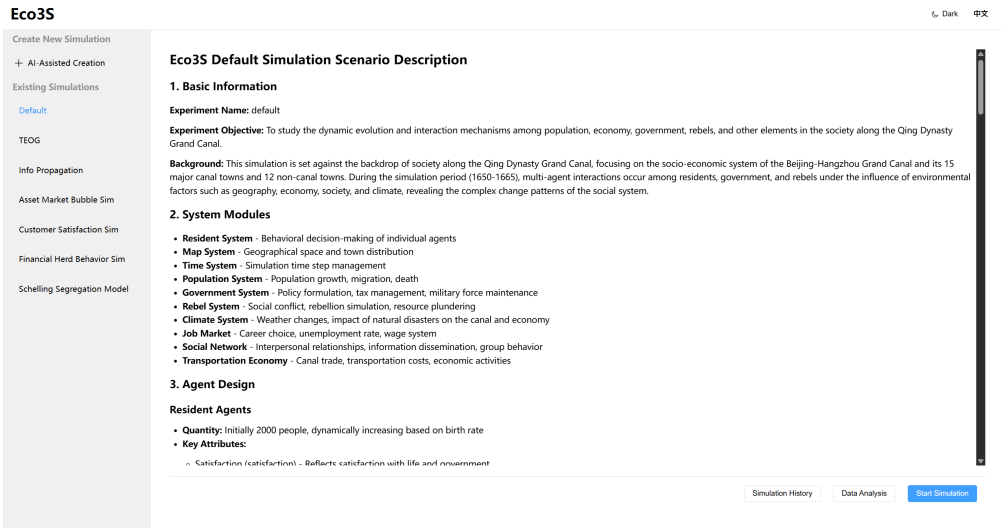
To facilitate the design, deployment, and analysis of large-scale social simulations, we developed a comprehensive web-based platform. This interface bridges high-level research objectives with low-level agent logic through an AI-assisted orchestration pipeline.

D.1. System Architecture and Functional Modules

The platform is built on a decoupled architecture with a **Vue 3** and **Element Plus** frontend and a **Flask RESTful API** backend. The primary interface is divided into a persistent navigation sidebar and a centralized workspace.

- **Navigation Sidebar:** Provides access to the AI-Assisted Creation suite and a repository of existing simulation projects.
- **Integrated Workspace:** Upon selecting a project, the workspace enables users to switch between four core views: *Scenario Description*, *Simulation Execution*, *Historical Records*, and *Data Analytics*.As shown in Figure 8, the interface provides a clear separation between navigation and workspace.

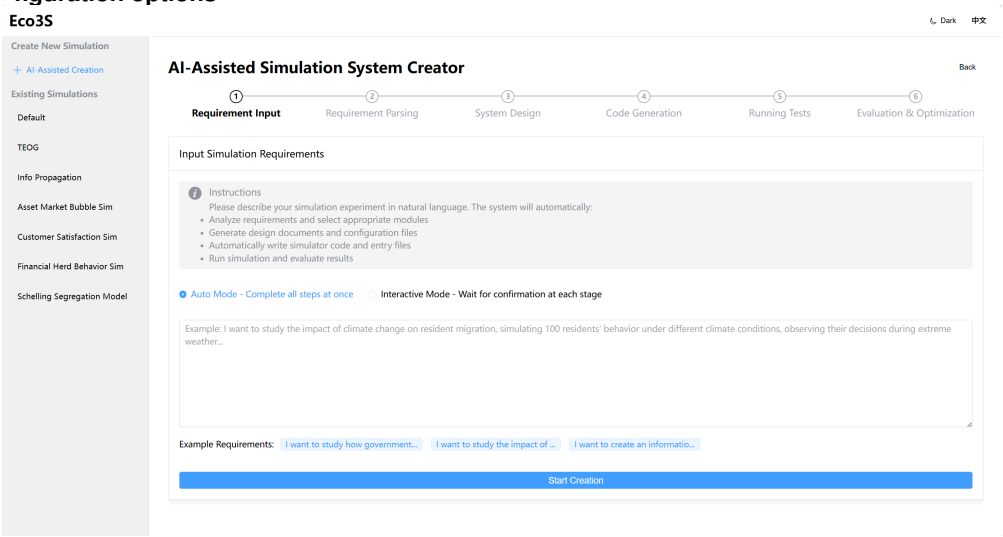
Figure 8 **Primary System Dashboard showing navigation and workspace layout**



D.2. AI-Assisted Lifecycle Orchestration

The core innovation of the platform is the automated generation pipeline, which orchestrates multiple LLM-based agents to transition through six distinct development phases.Users initiate this process through the AI-Assisted Creation interface shown in Figure 9.

Figure 9 AI-Assisted Simulation System Creator: Initial creation interface showing scenario setup and configuration options



The creation pipeline proceeds through seven distinct phases:

1. **Requirement Input:** Capturing natural language descriptions from users.
2. **Requirement Analysis:** Parsing and formalizing simulation objectives and constraints.
3. **System Design:** Defining the environment, agent roles, and interaction rules.
4. **Code Synthesis:** Generating the executable logic and scripts for the simulation engine.
5. **Execution Testing:** Deploying the environment and running initial validation.
6. **Evaluation and Optimization:** Performing comprehensive analysis and iterative refinement.

Users can toggle between an *Automated Mode* for rapid prototyping and an *Interactive Mode*, which allows for real-time human feedback verification and refinement at the end of each phase.The orchestration progress is monitored through the real-time interface depicted in Figure 10.

D.3. Execution, Configuration, and Real-Time Monitoring

Once a simulation scenario is synthesized, the system provides a structured environment for execution and parameter tuning.

- **Scenario Documentation:** Each project maintains a Markdown-based document detailing the experimental background, agent definitions, and objectives.
- **Interactive Configuration:** Users can fine-tune parameters—such as population size ($N_{residents}$), simulation duration (T_{steps}), and labor market mechanisms—through a syntax-highlighted YAML editor displayed concurrently with the execution view.

Figure 10 **AI Orchestration Pipeline Runtime Interface: Phase progress tracking, agent coordination status, and real-time execution logs**

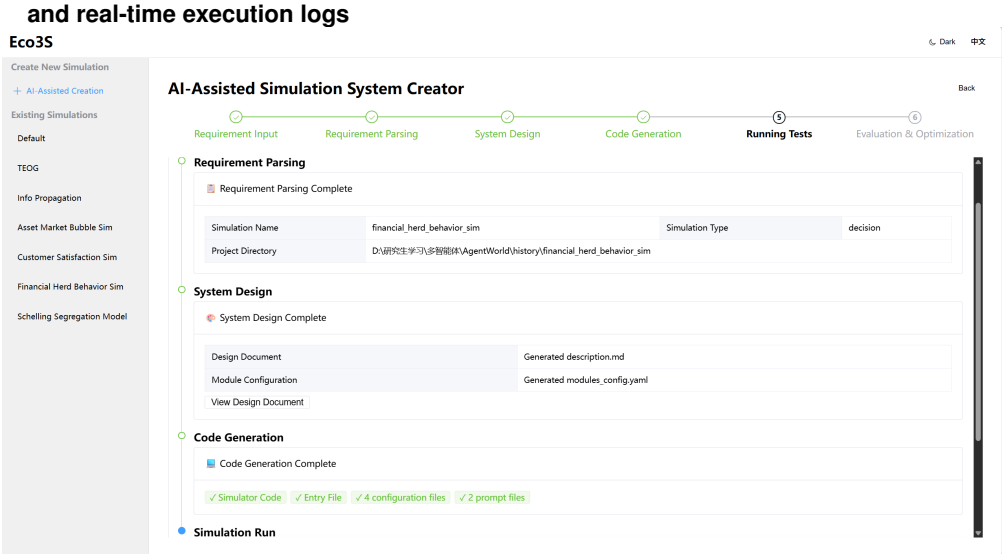


Figure 11 **Simulation Execution Interface with real-time telemetry and visualization**

- **Dynamic Monitoring:** During execution, the platform provides a synchronized view of simulation steps, real-time event logs, and live-updated charts (driven by Chart.js) visualizing key metrics such as economic indicators and agent satisfaction levels.Figure 11 demonstrates the real-time monitoring capabilities of the execution interface.

D.4. Historical Persistence and Analytics

To support iterative scientific inquiry, the system maintains a comprehensive data persistence layer.

- **Historical Repository:** Every simulation run is archived with its corresponding timestamps, configuration snapshots, and full execution logs. As shown in Figure , researchers can browse simulation records, including detailed log files and result visualizations from previous experiments.
- **Analytical Module:** The data analytics component aggregates and statistically analyzes multiple simulation runs of the same scale. By integrating data from identical experimental setups, researchers can obtain scientifically robust

Figure 12 **Experiment Analysis Interface: Configuring analysis parameters and viewing comparative results**

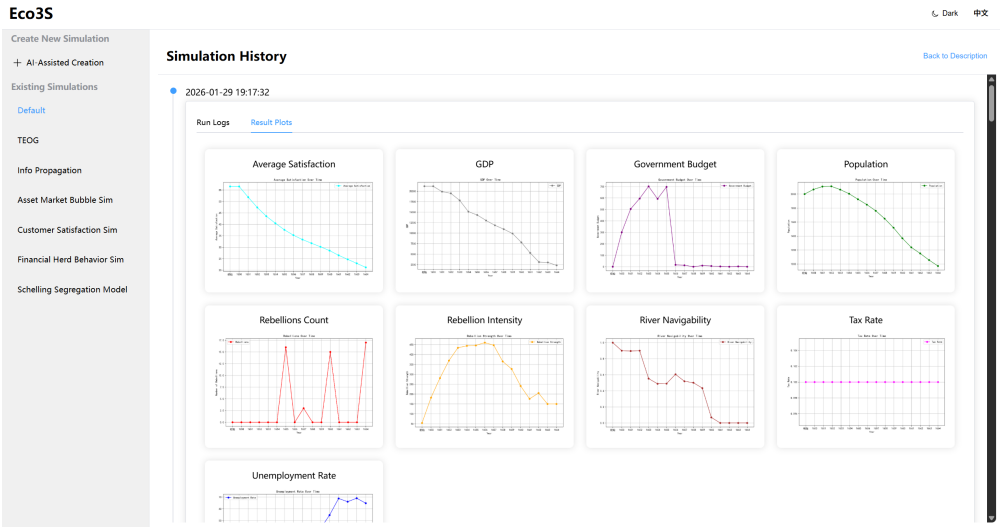
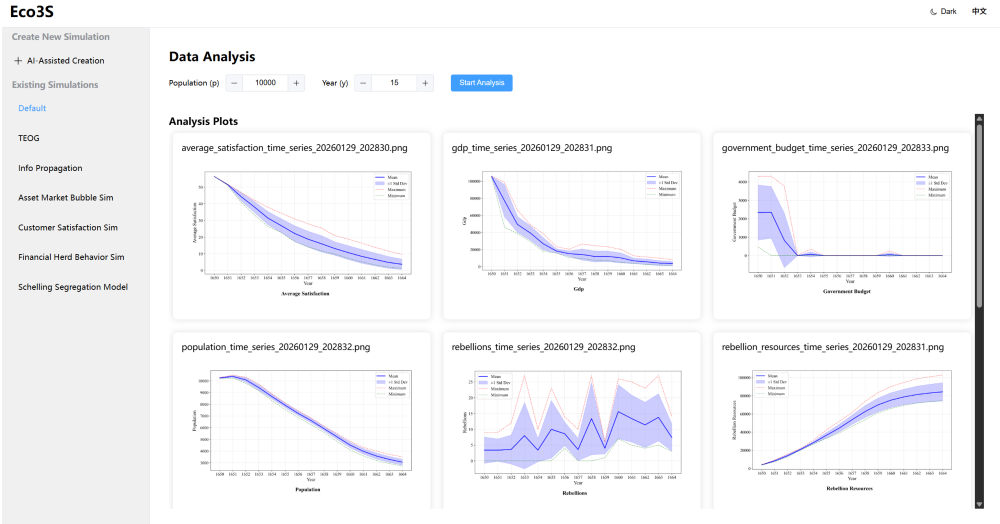


Figure 13 **Simulation Execution Interface with real-time telemetry and visualization**



statistical results. Parameters such as target year range and population size can be specified to generate comprehensive statistical reports (Figure 13).

D.5. Parameter Specifications

The operational logic of the simulation is governed by several core configuration manifests, summarized in Table 1.

Appendix E: System Dynamics Model Architecture

This appendix provides the detailed architecture and simulation results of the System Dynamics (SD) model, built via Vensim PLE, which serves as the benchmark for the comparative analysis in Section **??**.

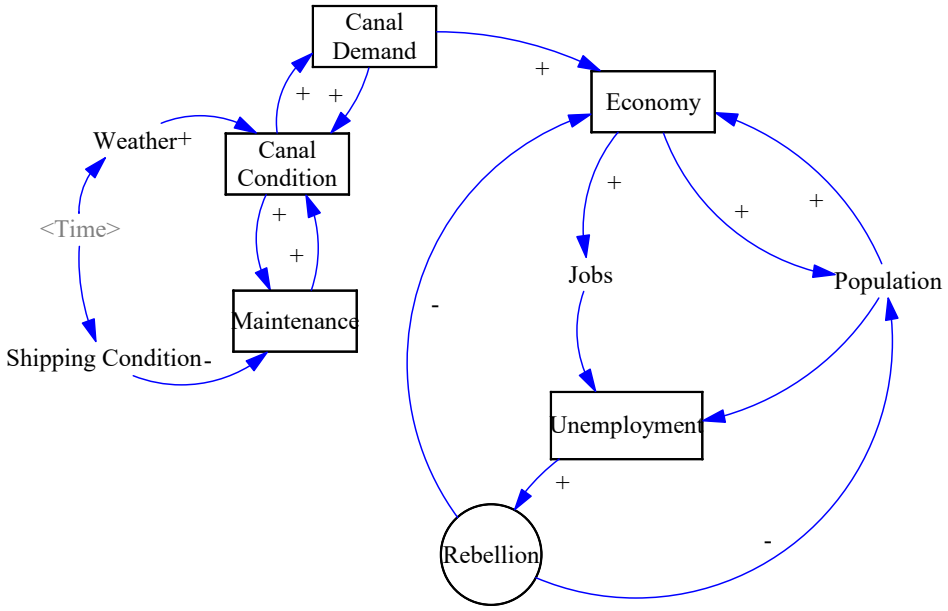
Figure 14 illustrates the structural design of the SD model, depicting the causal loops and stocks-and-flows used to represent the ”Rebel on the Canal” scenario. To evaluate the performance of this traditional approach, we executed the

Table 1 Primary Configuration Manifests and Descriptions

Manifest File	Functional Description
simulation_config.yaml	Defines population size, simulation steps, and initial environmental states.
jobs_config.yaml	Specifies occupational categories, wage structures, and employment transition rules.
resident_actions.yaml	Outlines the action space and trigger conditions for individual agents.
residents_prompts.yaml	Contains the LLM instructions for resident agent reasoning and social interaction logic.
government_prompts.yaml	Defines the administrative logic and fiscal policy-making instructions for governance agents.

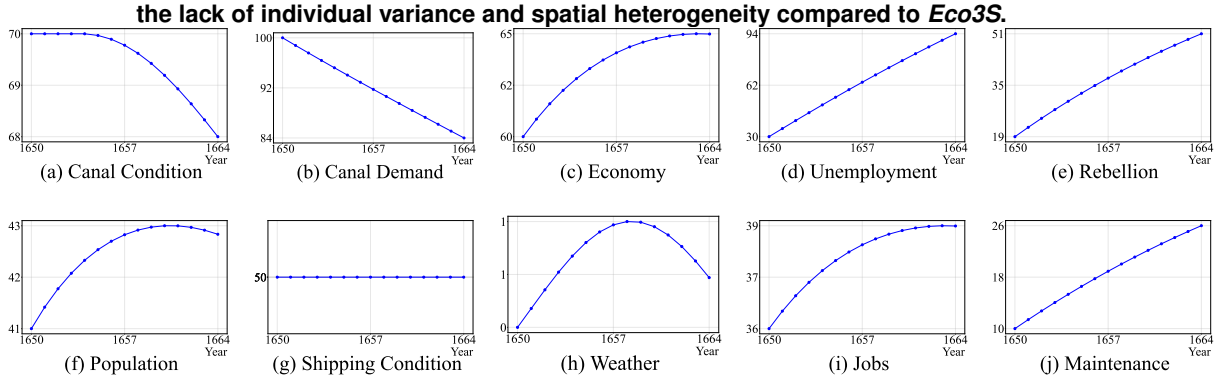
simulation under the same historical parameters as the Eco3S experiments. The resulting aggregate trends are presented in Figure .

Figure 14 System Dynamics Model Structure: Rebel on the Canal



As shown in Figure 15, While the SD model successfully captures the general macro-level trends—such as the steady decline in canal conditions leading to increased unemployment and social unrest—it exhibits several fundamental limitations when compared to the agent-based approach of *Eco3S*:

Figure 15 Vensim PLE Simulation Results (Traditional Benchmark). The smooth, aggregate curves highlight the lack of individual variance and spatial heterogeneity compared to *Eco3S*.



Lack of Individual Heterogeneity: Traditional SD models rely on aggregate stocks and flows, treating the population as a homogeneous mass. In contrast, *Eco3S* simulates thousands of unique agents with diverse personalities, income levels, and social ties, allowing for the emergence of complex behaviors that aggregate models cannot capture.

Absence of Spatio-Temporal Heterogeneity: As shown in Figure 15, the output curves are perfectly smooth, representing global averages. Traditional methods struggle to reflect spatial disparities, such as the specific geographic advantage of canal-adjacent towns versus inland regions. In contrast, the Delicate Environment Design of *Eco3S*—which integrates a dynamically evolving physical environment with a map-based heterogeneous information network—successfully captures how localized environmental decay triggers regional social instability.

Rigid Decision-making Mechanisms: In Vensim, interactions are governed by fixed mathematical equations. This approach lacks the nuanced, context-sensitive reasoning provided by the LLM-based decision-making in *Eco3S*, which can simulate sophisticated social strategies like propaganda dissemination and deliberative governance.

Appendix F: Robustness Analysis

To verify the robustness and flexibility of our framework, we conducted experiments using four different Large Language Model (LLM) providers as the cognitive engines for the agents:

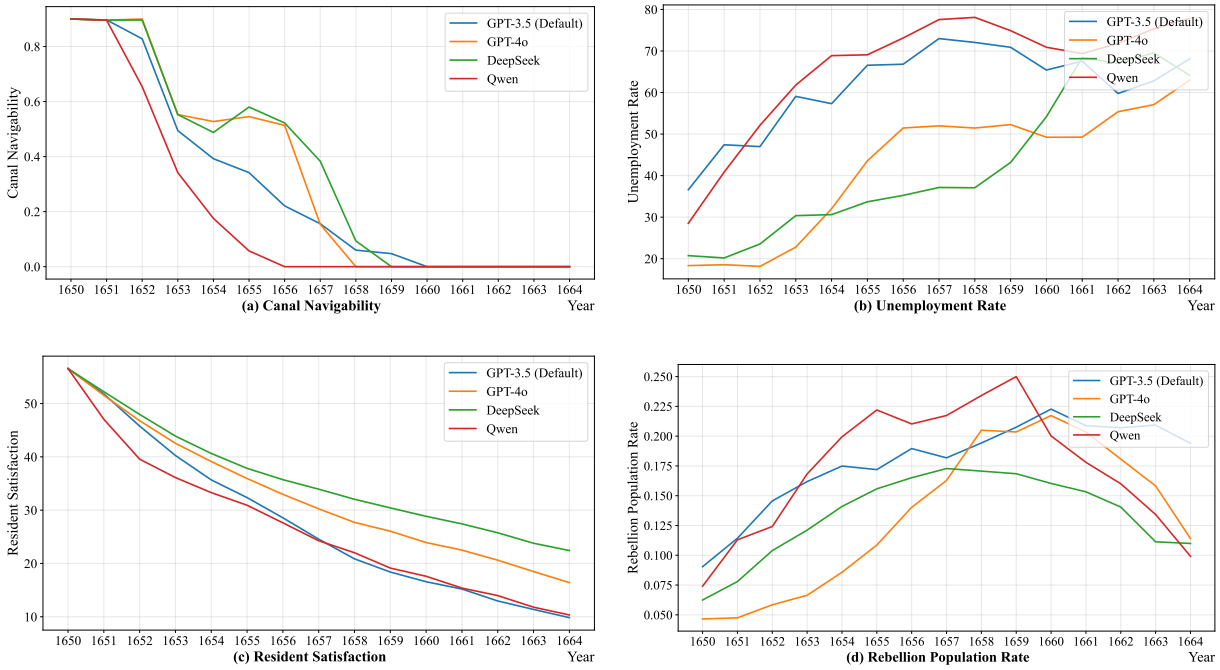
- **OpenAI GPT-3.5 Turbo** (gpt-3.5-turbo)
- **OpenAI GPT-4o** (gpt-4o)
- **DeepSeek** (deepseek-chat)
- **Qwen** (qwen3-235b-a22b)

For each provider, we ran independent simulation trials with identical initial conditions:

- Population size: $N = 2000$ residents
- Simulation duration: $T = 15$ steps (years)
- Economic parameters: Identical job distributions, wage structures, and market rules
- Social dynamics: Same interaction probability matrices and agent behavior templates
- Random seeds: **Stochastically assigned for each trial** to evaluate the framework's emergence under non-deterministic conditions rather than simple replication.

As illustrated in Figure 16, the simulation results across different LLMs exhibit a high degree of qualitative consistency. Specifically, Figure 16(a) shows that all models successfully capture the inevitable collapse of river

Figure 16 Consistency of simulation trends across different API providers



navigability due to siltation. This physical decay directly triggers a sharp rise in the unemployment rate (Figure 16b) and a continuous decline in average social satisfaction (Figure 16c). The social unrest, represented by the rebellion strength ratio in Figure 16(d), consistently peaks as satisfaction hits critical lows before tailing off due to systemic exhaustion.

The convergence of these trajectories—despite the use of different API architectures and stochastic seeds—demonstrates that the emergent socio-economic phenomena are driven by the internal logic of our framework rather than the specific biases of a single LLM. This confirms that the **framework is highly stable and remains unaffected by the choice of API provider**.

Appendix G: Empirical Baseline and Interpretation Guide

Table 2 presents the baseline Difference-in-Differences (DID) estimates from the study by Cao and Chen (2022), which examines the impact of the Grand Canal's abandonment on social conflict.

Data and Variable Definitions:

- **Dependent Variable (Rebellions):** The frequency of rebellion onsets per county, normalized by the initial population size (1600 population) and transformed using the inverse hyperbolic sine (arcsinh) for statistical consistency.
- **AlongCanal × Post:** This is the key interaction term. *AlongCanal* is a dummy variable (1 for canal-adjacent counties), and *Post* is a dummy variable (1 for years 1826 and after). The coefficient represents the treatment effect of canal abandonment.
- **Model Specifications (Columns 1–5):** The model progressively adds controls to ensure robustness. *Column 5* is the most stringent, controlling for:

1. **County & Year Fixed Effects (FE):** Accounting for time-invariant local factors and common annual shocks.

Table 2 Empirical Baseline: Impact of Canal Abandonment on Rebellions (?)

	Rebellions				
	(1)	(2)	(3)	(4)	(5)
AlongCanal × Post	0.0380	0.0369	0.0453	0.0427	0.0340
	(0.0166)	(0.0172)	(0.0173)	(0.0172)	(0.0166)
	[0.0165]	[0.0167]	[0.0167]	[0.0168]	[0.0168]
County & Year FE	Yes	Yes	Yes	Yes	Yes
Pre-reform Rebellion × Year	No	Yes	Yes	Yes	Yes
Province × Year FE	No	No	Yes	Yes	Yes
Prefecture Year Trend	No	No	No	Yes	Yes
Controls × Post	No	No	No	No	Yes
Mean of Dep. Var.	0.0330	0.0330	0.0330	0.0330	0.0330
Number of Observations	140,432	140,432	140,432	140,432	140,432
Number of Counties	536	536	536	536	536

Note: AlongCanal and Post are dummy variables that equal one for counties containing a stretch of the Grand Canal and for years following its abandonment, respectively, and zero otherwise. Standard errors in parentheses are clustered at the county level; Standard errors in brackets are Conley standard errors robust for spatial correlation. All coefficients for "AlongCanal × Post" are significant at the 5% level.

2. **Pre-reform Rebellion × Year:** Controlling for the possibility that counties already prone to unrest reacted differently.

3. **Province × Year FE & Prefecture Trends:** Controlling for region-specific socio-economic shifts over time.

4. **Controls × Post:** Controlling for geography (land size, ruggedness), demography (population density), and agriculture (crop suitability).

- **Statistical Notations:** Values in $(\cdot)$ are standard errors clustered at the county level. Values in $[\cdot]$ are Conley standard errors robust to spatial correlation within a 500km radius.

Calculation of Relative Increase in Rebellions:

To align our simulation results with the empirical findings of Cao and Chen (**?**), we calculate the relative increase in rebellions implied by their most stringent model (Column 5 of Table 2). Given the estimated coefficient 0.0340 and the sample mean of 0.0330, the calculation is as follows:

- The absolute change in the transformed dependent variable is: $\exp(0.0340)-1 \approx 0.0346$.
- The relative increase compared to the sample mean is: $0.0346/0.0330 \approx 1.05$.

This derived 105% relative increase serves as the quantitative benchmark for validating our simulation's conflict dynamics.

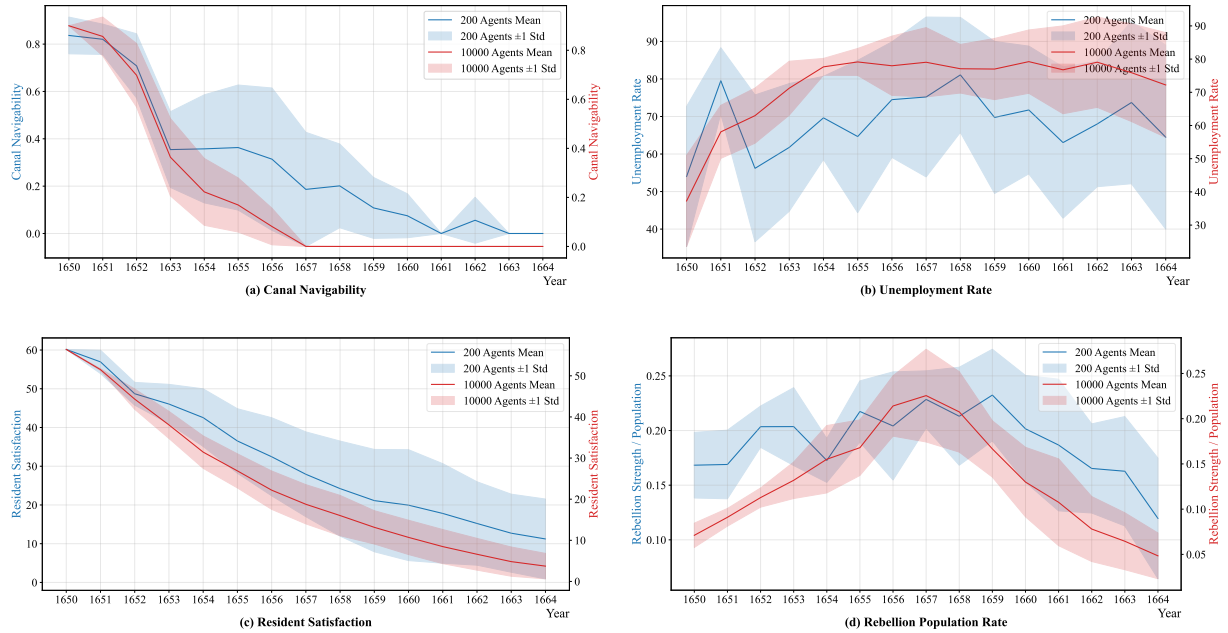
Appendix H: System Performance Details

H.1. Scaling Effect Analysis

We investigate the impact of agent population size by comparing two configurations: 200 versus 10,000 agents. As shown in Figure 17, simulations with 10,000 agents exhibit reduced fluctuations across all four sub-figures, as evidenced by consistently lower variance compared to the 200-agent case. Moreover, the larger agent scale yields both qualitative and quantitative differences in the observed outcomes. For example, in sub-figure (a), the 10,000-agent simulation terminates earlier and with markedly lower variance. Sub-figure (d) reveals a distinct transition occurring in 1657 under

the 10,000-agent setting—an event that aligns precisely with the termination time observed in sub-figure (a). This correspondence may be attributed to the closure of the canal, which triggered substantial population outflow—primarily among communities residing along the canal and dependent on it for livelihood—thereby reducing the pool of potential participants in the rebellion. Collectively, these results indicate that increasing the agent scale enhances the robustness and clarity of the simulation outcomes, supporting more reliable and discernible conclusions.

Figure 17 Comparison Between Different Agent Scale



H.2. Efficiency Benchmarks

We evaluate the computational efficiency of the *Eco3S* framework by measuring runtime under varying simulation configurations, particularly with respect to the number of agents and simulation steps. As summarized in Table 3, execution time scales sublinearly with increases in both agent count and temporal horizon, demonstrating the framework's suitability for large-scale socio-economic modeling. For instance, simulating the Canal Decay and Origins of Governance scenarios with 2,000 agents over 10 steps takes 30:56.2 and 39:05.1 minutes, respectively. When the number of steps is doubled to 20 (with agent count held constant at 2,000), runtime increases modestly to 33:05.8 and 59:06.0 minutes—reflecting the relatively low per-step computational overhead once the agent population is initialized. By contrast, reducing the agent count to 500 (with 10 steps) yields runtime of 10:53.7 and 20:41.2 minutes—more than a quarter of the time required for 2,000 agents. This subproportional scaling arises because *Eco3S* leverages concurrent agent processing: agent decisions are computed in parallel wherever possible, minimizing sequential bottlenecks in the simulation loop.

Notably, the dominant constraint on scalability in practice is not the framework's architecture, but external LLM platform rate limits—which govern the throughput of language model queries.

Table 3 Comparison between different simulation scales

Agents	Steps	Canal Decay	Origins of Governance
100	5	02:18.8	01:33.7
	10	06:03.7	03:04.9
	20	09:33.9	18:36.9
500	5	03:51.0	05:05.6
	10	10:53.7	20:41.2
	20	12:28.4	34:06.5
2000	5	15:33.3	18:19.3
	10	30:56.2	39:05.1
	20	33:05.8	59:06.0
10000	5	1:22:50	-